

Unit 4: Methods of Quantum Mechanics:

Classical mechanics or Newtonian mechanics is applicable only in case of study of the behavior of macroscopic world in which body consider as mass and follow Newton's law. This mechanics is insufficient to address the study of particles in microscopic (atomic) scale so, a new mechanics named Quantum mechanics is in practice which is based on particle as a wave and follow Schrodinger's wave equation. Quantum mechanics is indeterministic and is probabilistic in nature.

In 1924 A.D Erwin Schrodinger, consider a number of plane waves in wave packet form in order to describe the behavior of matter wave. In general, it is expressed as i.e.

$$\begin{aligned}\psi(x, t) &= A\cos(\omega t - kx) + [-iA\sin(\omega t - kx)] \\ \text{or, } \psi(x, t) &= A\cos(\omega t - kx) - iA\sin(\omega t - kx) \\ \text{or, } \psi(x, t) &= Ae^{-i(\omega t - kx)} \dots\dots\dots 1\end{aligned}$$

$$\text{Further; } E = \hbar\omega \text{ or, } \omega = \frac{E}{\hbar} \text{ and } p = \hbar k \text{ or, } k = \frac{p}{\hbar}$$

Then with the equation 1 becomes;

$\psi(x, t) = Ae^{\frac{-i}{\hbar}(Et - px)}$ which is the matter wave equation representing the particle in space with finite energy and momentum also known as wavefunction.

#Probability of finding particle:

According to Born's statistical interpretation, a probability of a finding a particle in space over given interval is given by;

$$\text{probability}(p) = \int_a^b |\psi(x, t)|^2 dx \text{ where } \Psi(x, t) \text{ is wave function representing a particle at 'x' position in time 't' in one-dimensional space.}$$

$$\text{or, } \text{probability}(p) = \int_a^b \psi^*(x, t)\psi(x, t)dx$$

Thus, the square integration of wave function over given intervals is define as the probability of being finding a particle in given space in Quantum mechanics. Quantum mechanics doesn't give the prediction of where the particle will be located exactly at next moment in space however it gives an idea about more likely hood of finding particle in space. If we made a measurement then there will be probability of finding particle everywhere in space but the position at which amplitude of wave is maximum represent the most probable site for finding a particle.

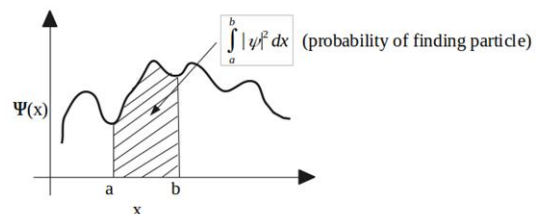
For a bounded particle in space the probability of finding particle in given interval is unit while its value outside the interval is zero i.e.

$$\begin{aligned}p &= \int_a^b |\psi(x, t)|^2 dx = 1 \text{ for } a \leq x \leq b \text{ and} \\ \text{or, } p &= \int_a^b |\psi(x, t)|^2 dx = 0 \text{ for } x \leq a; x \geq b\end{aligned}$$

Further, for free particle in space; its probability is unit

$$\text{probability}(p) = \int_{-\infty}^{\infty} |\psi(x, t)|^2 dx = 1$$

Which predicts the certainty of particle in space.



Normalization of wave function and its meaning:

The solution obtained from Schrodinger wave equation, may always don't have a proper physical sense. If particle have an existence in space over range then its probability has unit value i.e.

$$\text{probability}(p) = \int_{-\infty}^{\infty} |\psi(x.t)|^2 dx = 1$$

In certain cases, thus obtained wave function don't support above condition hence a process is to be carried to valid the condition, in which a wave function is simply multiplied by some constant such that it agrees the above relation.

Thus, the process of multiplying a wave function by a suitable real number constant such that its square integrable over an entire space equal to unit is called normalization of wave function & the constant is called normalization constant. The condition $\int_{-\infty}^{\infty} A^2 |\psi(x.t)|^2 dx = 1$ is called normalization condition.

Hence, the normalization process of wave function conforms the existence of particle in space somewhere and hence provide the physical sense to a wave function obtained from Schrodinger wave equation.

Characteristics of wave function (interpretation of wave function or Physical meaning of wave function):

A wave function is a mathematical complex function obtained by solving Schrodinger wave equation, that contains the information about particle but doesn't have physical existence that is to say it has no more sense of physical quantity. In order to have meaningful sense to the physical world it should have following properties or characteristics,

1. The wave function must be finite, single valued and continuous in space.
2. The derivative of wave function must be existed, finite, single valued and continuous.
3. The wave function must be normalized i.e. the square integration of wave function over space should be unit. This gives the physical sense of certainty for existence of particle in space.

$$\text{probability}(p) = \int_{-\infty}^{\infty} |\psi(x.t)|^2 dx = 1$$

4. For bounded system its wave function should be zero for 'x' tends to infinity.
5. The wave function should satisfied a boundary conditions in case of bounded system.

Schrodinger Wave equation:

Schrodinger wave equation is the fundamental equation of quantum mechanics which is second order partial differential equation over the variable ' Ψ ' i.e. in wave function of particle. The wave function in quantum mechanics includes all the information about particle's position, velocity, momentum and energy. Schrodinger wave equation has the equivalent role in quantum mechanics as the newton's equation for body in classical or Newtonian mechanics. Schrodinger's wave equation are of two types; time independent Schrodinger wave equation and time dependent Schrodinger wave equation.

1. Time dependent Schrodinger wave equation:

For a wave moving with velocity 'v' in a x-space at given time 't' and producing 'y' displacement in particles then the equation of wave can be written as;

$$\frac{\partial^2 y}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 y}{\partial t^2} \dots\dots\dots 1$$

whose solution in general comes as $y = A_0 e^{i(kx - \omega t)}$; which contains real and imaginary part.

For quantum mechanical system with energy 'E' and momentum 'p' the wave associated with it is represented by a wave function just like the solution of wave equation 1. and given as

$$\psi(x,t) = A_0 e^{i(kx - \omega t)} \dots\dots\dots 2$$

Further before deriving Schrodinger wave equation, it is assumed that the wave equation or quantum particle obeying following principle

1. De-Broglie hypothesis and plank- Einstein postulates:

$$\lambda = \frac{h}{p} \text{ or, } p = \frac{h}{\lambda} = \frac{h}{2\pi} \frac{2\pi}{\lambda} = \hbar k \quad \text{or, } k = \frac{p}{\hbar} \quad \text{and}$$

$$E = \hbar \omega \quad \text{or, } E = \frac{h}{2\pi} 2\pi \omega = \hbar \omega \quad \text{or, } \omega = \frac{E}{\hbar}$$

2. Energy conservation principle:

The solution of wave equation i.e. wave function of quantum mechanical system should follow conservation of energy. The total energy of system is equal to the sum of its kinetic energy and potential energy i.e.

$$E = K.E + p.E$$

$$E = \frac{p^2}{2m} + V(x, t) \quad \text{for conservative system } F = -\frac{dV}{dx}$$

3. Principle of linearity superposition:

For the wave equation having number of solution then the liner combination of these solutions should be also a solution of Schrodinger wave equation. i.e.

$$\psi = C_1\psi_1 + C_2\psi_2$$

4. Force for conservative system is zero i.e. at constant potential region

$$F = -\frac{dV}{dx} = 0 \quad \text{and quantum system becomes free particle.}$$

Following all above assumptions, wave function from equation 2 becomes;

$$\psi(x, t) = A_0 e^{\frac{i}{\hbar}(px - Et)} \dots\dots\dots 3$$

Differentiating equation 3 partially with respect to 'x'

$$\frac{\partial \psi(x, t)}{\partial x} = A_0 \frac{ip}{\hbar} e^{\frac{i}{\hbar}(px - Et)}$$

Again Differentiating partially with respect to 'x' for above eqn.

$$\frac{\partial^2 \psi(x, t)}{\partial x^2} = A_0 \left(\frac{ip}{\hbar}\right)^2 e^{\frac{i}{\hbar}(px - Et)}$$

$$\text{or, } \frac{\partial^2 \psi(x, t)}{\partial x^2} = -A_0 \left(\frac{p}{\hbar}\right)^2 e^{\frac{i}{\hbar}(px - Et)}$$

$$\text{or, } \frac{\partial^2 \psi(x, t)}{\partial x^2} = -\left(\frac{p^2}{\hbar^2}\right) \psi(x, t) \quad [\text{from eqn 3}]$$

$$p^2 \psi(x, t) = -\hbar^2 \frac{\partial^2 \psi(x, t)}{\partial x^2} \dots\dots\dots 4$$

Further; differentiating equation 3 partially with respect to 't', we get;

$$\frac{\partial \psi(x, t)}{\partial t} = A_0 \frac{-iE}{\hbar} e^{\frac{i}{\hbar}(px - Et)}$$

$$\text{or, } \frac{\partial \psi(x, t)}{\partial t} = \frac{-iE}{\hbar} \psi(x, t)$$

$$\text{or, } E\psi(x, t) = \frac{-\hbar}{i} \frac{\partial \psi(x, t)}{\partial t}$$

$$E\psi(x, t) = i\hbar \frac{\partial \psi(x, t)}{\partial t} \dots\dots\dots 5$$

Now, applying principle of conservation of energy; we get

$$E = \frac{p^2}{2m} + V$$

multiplying both sides with wave function $\psi(x, t)$

$$E\psi(x, t) = \frac{p^2}{2m} \psi(x, t) + V\psi(x, t)$$

from equation 4 and 5; we have

$$i\hbar \frac{\partial \psi(x, t)}{\partial t} = \frac{-\hbar^2}{2m} \frac{\partial^2 \psi(x, t)}{\partial x^2} + V(x, t)\psi(x, t)$$

$$\text{or, } \frac{-\hbar^2}{2m} \frac{\partial^2 \psi(x, t)}{\partial x^2} + V(x, t)\psi(x, t) = i\hbar \frac{\partial \psi(x, t)}{\partial t}$$

This is required expression for the time dependent Schrodinger Wave equation in 1-dimension.

For three dimension it is written as;

$$\frac{-\hbar^2}{2m} \left[\frac{\partial^2 \psi(x, y, z, t)}{\partial x^2} + \frac{\partial^2 \psi(x, y, z, t)}{\partial y^2} + \frac{\partial^2 \psi(x, y, z, t)}{\partial z^2} \right] + V(x, y, z, t)\psi(x, y, z, t) = i\hbar \frac{\partial \psi(x, y, z, t)}{\partial t}$$

$$\text{or, } \frac{-\hbar^2}{2m} \left[\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) + V(x, y, z, t) \right] \psi(x, y, z, t) = i\hbar \frac{\partial \psi(x, y, z, t)}{\partial t}$$

$$\text{or, } \frac{-\hbar^2}{2m} [\nabla^2 + V(r, t)] \psi(r, t) = i\hbar \frac{\partial \psi(r, t)}{\partial t} \text{ where, } \nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \text{ is called Laplacian operator.}$$

2. Time independent Schrodinger wave equation:

In many problems potential energy of a particle depends on its position only i.e. $V(x, t) = V(x)$ which is independent of time. In such case, Schrodinger wave equation is only position dependent so, it is also known as steady state of Schrodinger wave equation.

The wave function representing particle hence can be expressed as;

$$\psi(x) = A_0 e^{\frac{ipx}{\hbar}} \dots\dots\dots 1$$

From the conservation of energy, The total energy of system is equal to the sum of its kinetic energy and potential energy i.e.

$$E = K.E + p.E$$

$$E = \frac{p^2}{2m} + V(x) \dots\dots\dots 2$$

Differentiating equation 1 with respect to 'x'

$$\frac{d\psi(x, t)}{dx} = A_0 \frac{ip}{\hbar} e^{\frac{ipx}{\hbar}}$$

Differentiating again with respect to 'x' for above eqn.

$$\frac{d^2 \psi(x, t)}{dx^2} = A_0 \left(\frac{ip}{\hbar} \right)^2 e^{\frac{ipx}{\hbar}}$$

$$\text{or, } \frac{d^2 \psi(x, t)}{dx^2} = -A_0 \left(\frac{p}{\hbar} \right)^2 e^{\frac{ipx}{\hbar}}$$

$$\text{or, } \frac{d^2 \psi(x, t)}{dx^2} = -\left(\frac{p^2}{\hbar^2} \right) \psi(x) \quad [\text{from eqn 1}]$$

$$p^2 \psi(x, t) = -\hbar^2 \frac{d^2 \psi(x, t)}{dx^2} \dots\dots\dots 3$$

Now, applying principle of conservation of energy; we get

$$E = \frac{p^2}{2m} + V$$

multiplying both sides with wave function $\psi(x)$

$$E\psi(x) = \frac{p^2}{2m} \psi(x) + V\psi(x)$$

using equation 3; we have

$$E\psi(x) = \frac{-\hbar^2}{2m} \frac{\partial^2 \psi(x,t)}{\partial x^2} + V\psi(x)$$

$$\text{or, } \frac{\hbar^2}{2m} \frac{\partial^2 \psi(x,t)}{\partial x^2} + [E - V(x)]\psi(x) \dots \dots \dots 4$$

$$\text{or, } \frac{d^2 \psi(x)}{dx^2} + \frac{2m}{\hbar^2} [E - V(x)]\psi(x) = 0$$

Which is required expression for the time independent Schrodinger wave equation for quantum system.

For three-dimension time independent Schrodinger wave equation is;

$$\nabla^2 \psi(x, y, z) + \frac{2m}{\hbar^2} [E - V(x, y, z)]\psi(x, y, z) = 0$$

$$\text{or, } \nabla^2 \psi(r) + \frac{2m}{\hbar^2} [E - V(r)]\psi(r) = 0 \quad \text{where, } \nabla^2 = \frac{d^2}{dx^2} + \frac{d^2}{dy^2} + \frac{d^2}{dz^2} \text{ is called Laplacian operator.}$$

Expectation values of Dynamical Quantities (Quantum system):-

The solution of this equation gives a complex mathematical wave function which includes a lot of information about particle but doesn't give a direct concrete physical meaning or value of any physical quantity.

For, large number of measurements made on similar independent quantum mechanical system that exist in given 'Ψ' state, then the mathematical average or mean value of such repeated measurements of a physical quantity over similar quantum system is called an expectation value. Hence, expectation value gives the most possible value of particular physical quantity in quantum mechanics.

The expectation value of any physical quantity f(r) in 3-dimension is given by;

$$\langle f \rangle = \int_{-\infty}^{\infty} P(r) f(r) dv \quad \text{where, } p(r) \text{ is probability value of particle in given volume.}$$

$$\text{Or, } \langle f \rangle = \int_{-\infty}^{\infty} \Psi^*(r, t) f(r) \Psi(r, t) d^3r \dots \dots \dots 1$$

where 'Ψ' is normalized wave function.

In case of not normalized wave function, expectation value of physical quantity f(r) is given by;

$$\langle f \rangle = \frac{\int_{-\infty}^{\infty} \Psi^*(r, t) f(r) \Psi(r, t) d^3r}{\int_{-\infty}^{\infty} \Psi^*(r, t) \Psi(r, t) d^3r} \dots \dots \dots 2$$

Note: - expectation value of f(x) in one dimension is

$$\langle f \rangle = \frac{\int_{-\infty}^{\infty} \Psi^*(x, t) f(x) \Psi(x, t) dx}{\int_{-\infty}^{\infty} \Psi^*(x, t) \Psi(x, t) dx}$$

-for e.g. expectation value of position 'x' in one dimension is

$$\langle x \rangle = \frac{\int_{-\infty}^{\infty} \Psi^*(x, t) x \Psi(x, t) dx}{\int_{-\infty}^{\infty} \Psi^*(x, t) \Psi(x, t) dx}$$

for normalized wave function; $\int_{-\infty}^{\infty} \Psi^*(x, t) \Psi(x, t) dx = 1$ so,

$$\langle x \rangle = \int_{-\infty}^{\infty} \Psi^*(x, t) x \Psi(x, t) dx$$

Operators in Quantum Mechanics:-

In quantum mechanics, a behavior of particle is defined by a Schrodinger wave equation. The solution obtained by solving these wave equations is just a mathematical complex function representing the particle and doesn't give the exact any physical quantities. So, the operation of wave function with suitable operator is required to obtain particular physical quantity.

An operator is a mathematical instruction which performs a mathematical operation in a wave packet (wave function) and result to some number as eigen value representing particular information about the physical system. Hence, the possible outcomes of measurement over quantum system is given by eigen value equation operated by an operator in quantum mechanics.

For e.g. the eigen value obtained with energy operator over wave function gives the possible energy levels of the system. Operators are very useful in quantum mechanics in order to get the expectation value of quantum mechanical system.

Note:

A eigen value equation contains an operator which when operate with a function result to some number times the wave function. For any operator $\hat{a}$ over ψ function and this result to λ eigen value then eigen value equation is,

$$\hat{a}\psi = \lambda\psi$$

Operators for position momentum and energy of quantum system.

A solution of Schrodinger wave equations gives the complex mathematical result known as matter wave function. The wave function representing quantum particle in one-dimensional space with finite energy and momentum in general form is given by,

$$\psi(x, t) = A e^{\frac{i}{\hbar}(px - Et)} \dots\dots\dots 1$$

Differentiating eqn 1 w.r.to. 'x'; we get,

$$\frac{\partial \psi(x, t)}{\partial x} = A \frac{ip}{\hbar} e^{\frac{i}{\hbar}(px - Et)}$$

$$\text{or, } \frac{\partial \psi(x, t)}{\partial x} = \frac{ip}{\hbar} \psi(x, t)$$

$$\text{or, } \frac{\hbar}{i} \frac{\partial \psi(x, t)}{\partial x} = p\psi(x, t)$$

$$\text{or, } (-i\hbar \frac{\partial}{\partial x})\psi(x, t) = p\psi(x, t)$$

$$\text{or, } \hat{p}\psi(x, t) = p\psi(x, t)$$

where, $\hat{p} = i\hbar \frac{\partial}{\partial x}$ is momentum operator in one dimension.

Again, Differentiating eqn 1 w.r.to. 't'; we get,

$$\frac{\partial \psi(x, t)}{\partial t} = -A \frac{iE}{\hbar} e^{\frac{i}{\hbar}(px - Et)}$$

$$\text{or, } \frac{\partial \psi(x, t)}{\partial t} = \frac{-iE}{\hbar} \psi(x, t)$$

$$\text{or, } \frac{-\hbar}{i} \frac{\partial \psi(x, t)}{\partial t} = E\psi(x, t)$$

$$\text{or, } (i\hbar \frac{\partial}{\partial t})\psi(x, t) = E\psi(x, t)$$

$$\text{or, } \hat{H}\psi(x, t) = E\psi(x, t) \text{ where, } \hat{H} = i\hbar \frac{\partial}{\partial t} \text{ is an total energy operator.}$$

Application of Schrodinger Theory :-

I. Particle (A free electron) in 1-Dimensional infinite potential well:-

Let us, consider a free electron of mass 'm' confined in a infinite potential well of length 'a' define as

$$V = 0 \quad ; \quad 0 \leq x \leq a$$

$$= \infty \quad ; \quad x < 0 \text{ and } x > a$$

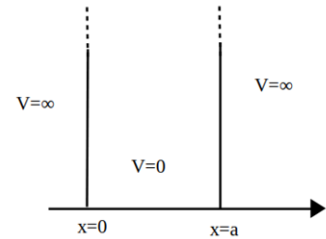


Fig: one dimensional infinite potential well.

The time independent Schrodinger wave equation for a free electron with wave function $\Psi_n(x)$ with energy E_n energy eigen value at n^{th} energy level is given by.

$$\hat{H}\psi_n(x) = E_n\psi_n(x)$$

$$\text{or, } \frac{-\hbar^2}{2m} \frac{d^2}{dx^2} \psi_n(x) = E_n\psi_n(x) \quad [\text{being } V(x) = 0 \text{ for } 0 \leq x \leq a]$$

$$\text{or, } \frac{d^2}{dx^2} \psi_n(x) + \frac{2mE_n}{\hbar^2} \psi_n(x) = 0$$

$$\text{or, } \frac{d^2}{dx^2} \psi_n(x) + k^2 \psi_n(x) = 0 \dots\dots\dots 1$$

$$\text{where, } k^2 = \frac{2mE_n}{\hbar^2} \dots\dots\dots 2 \quad \text{is a constant called momentum wave}$$

vector of wave of electron.

The general solution of equation 1 can be written as;

$$\psi_n(x) = A\cos Kx + B\sin Kx \dots\dots\dots 3$$

Using the boundary condition that the wave is absent at wall of box

$$\text{i.e. } \psi_n(0) = 0 \text{ at } x=0 \text{ and}$$

$$\psi_n(a) = 0 \text{ at } x=a,$$

we get;

$$\psi_n(0) = 0$$

$$\text{or, } A\cos 0 + B\sin 0 = 0$$

$$\text{or, } A + 0 = 0$$

$$\text{or, } A = 0$$

Then solution eqn 3 becomes;

$$\psi_n(x) = B\sin Kx \dots\dots\dots 4$$

Again, applying 2nd boundary condition;

$$\psi_n(a) = 0$$

$$\text{or, } B\sin ka = 0$$

$$\text{or, } \sin ka = 0$$

$$\text{or, } \sin ka = \sin n\pi \quad \text{where, } n=0,1,2,3,\dots\dots\dots \text{ is an integer.}$$

$$k = \frac{n\pi}{a} \dots\dots\dots 5$$

Now, the solution of wave function of an electron in 1-D potential box becomes;

$$\psi_n(x) = B \sin \frac{n\pi}{a} x \dots\dots\dots 6$$

where B is constant which can be determined by applying Normalization condition.

We know, the normalization condition for wave function is;

$$\text{probability}(p) = \int_{-\infty}^{\infty} |\psi(x, t)|^2 dx = 1$$

As particle is confined inside box of 'a' width, so wave function outside $0 < x < a$ region vanishes.

So,

$$\text{or, } \int_0^a \psi^*(x, t) \psi(x, t) dx = 1$$

$$\text{or, } \int_0^a B \sin \frac{n\pi}{a} x B \sin \frac{n\pi}{a} x dx = 1$$

$$\text{or, } B^2 \int_0^a \sin^2 \frac{n\pi}{a} x dx = 1$$

$$\text{or, } B^2 \int_0^a \frac{(1 - \cos \frac{2n\pi}{a} x)}{2} dx = 1$$

$$\text{or, } \frac{B^2}{2} \left[\int_0^a dx - \int_0^a \cos \frac{2n\pi}{a} x dx \right] = 1$$

$$\text{or, } \frac{B^2}{2} \left[\int_0^a dx - \int_0^a \cos \frac{2n\pi}{a} x dx \right] = 1$$

$$\text{or, } \frac{B^2}{2} \left[a - \frac{\sin \frac{2n\pi}{a} x}{\frac{2n\pi}{a}} \right] = 1$$

$$\text{or, } \frac{B^2}{2} \left[a - \frac{a}{2n\pi} \sin \frac{2n\pi}{a} x \right] = 1$$

$$\text{or, } \frac{B^2}{2} \left[a - \frac{a}{2n\pi} (\sin \frac{2n\pi}{a} a - \sin 0) \right] = 1$$

$$\text{or, } \frac{B^2 a}{2} = 1$$

$$\text{or, } B = \sqrt{\frac{2}{a}}$$

So, the normalized wave function that representing electron inside one-dimensional potential well of 'a' width becomes,

$$\psi_n(x) = \sqrt{\frac{2}{a}} \sin \frac{n\pi}{a} x \dots\dots\dots 7$$

Further, From eqn 2 and 5; the energy of particle in one-dimensional potential well is

$$\left(\frac{n\pi}{a}\right)^2 = \frac{2mE_n}{\hbar^2}$$

$$\text{or, } E_n = \frac{n^2 \hbar^2 \pi^2}{2ma^2} \dots\dots\dots 7$$

This is the required expression of energy of n^{th} level occupied by an electron in 1-D infinite potential well or box. This clearly states that the energy of an electron inside a box is quantized i.e. possible values of energies for electron are discrete values only.

Hydrogen Atoms Problem: -

A hydrogen atom consists of one proton about its nucleus with an orbital electron revolving around this nucleus in fixed orbit. An electron in its orbit is bounded to nucleus with an attractive coulomb potential energy

$$V(r) = \frac{-e^2}{4\pi\epsilon_0 r} \dots\dots\dots 1$$

In quantum mechanics, the orbital electron is assumed to be equivalent to wave system bounded in above potential well.

Further for m_p and m_e be the mass of proton and electron respectively then reduced mass of an atom is given by,

$$\mu = \frac{\mu_p \mu_e}{\mu_p + \mu_e} \dots\dots\dots 2$$

We know , the three-dimensional time independent Schrodinger wave equation is given by,

$$\nabla^2 \psi(x, y, z) + \frac{2m}{\hbar^2} [E - V(r)]\psi(x, y, z) = 0 \dots\dots\dots 3$$

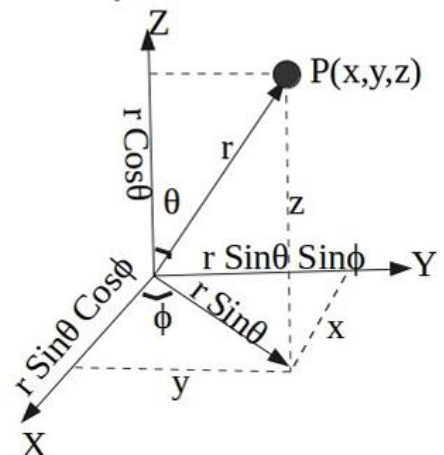
where, $\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$ is called Laplacian operator in cartesian co-ordinate system.

So, for the wave function of an electron; mass is replace by reduced mass and equation 3 becomes,

$$\nabla^2 \psi(x, y, z) + \frac{2\mu}{\hbar^2} [E - V(r)]\psi(x, y, z) = 0 \dots\dots\dots 4$$

Since the potential for an electron defined by equation 1 is symmetric in nature about the nucleus so, it is easier to solve above equation in polar co-ordinate (r, θ , ϕ) system. The relation between variables of cartesian coordinate system and polar co-ordinate system are

$$\left. \begin{aligned} x &= r \sin \theta \sin \phi \\ y &= r \sin \theta \cos \phi \\ z &= r \cos \theta \end{aligned} \right\} \dots\dots\dots 5$$



then, the Laplacian operator in three-dimension in polar co-ordinate becomes,

$$\nabla^2 = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 \frac{\partial}{\partial r}) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta \frac{\partial}{\partial \theta}) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \dots\dots\dots 6$$

From equation 4 and 6;

$$\left(\frac{1}{r^2} \frac{\partial}{\partial r} (r^2 \frac{\partial \psi(r, \theta, \phi)}{\partial r}) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta \frac{\partial \psi(r, \theta, \phi)}{\partial \theta}) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 \psi(r, \theta, \phi)}{\partial \phi^2} \right) + \frac{2\mu}{\hbar^2} [E - V(r)]\psi(r, \theta, \phi) = 0 \dots\dots\dots 7$$

As wave function Ψ is a function of three independent variables r, θ and ϕ so, using separation of variable method to solve equation 7, We can write;

$$\psi(r, \theta, \phi) = R(r)\Theta(\theta)\Phi(\phi) \dots\dots\dots 8$$

here, R refers to Radial wave function of an electron of H-atom,

Θ refers to orbital wave function of an electron of H-atom,

Φ refers to azimuthal wave function of an electron of H-atom .

and orbital along with azimuthal wave functions collectively known as polar wave function.

using eqn 8 in 7; we get,

$$\frac{\Theta\Phi}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial R}{\partial r} \right) + \frac{R\Phi}{r^2 \sin\theta} \frac{\partial}{\partial \theta} \left(\sin\theta \frac{\partial \Theta}{\partial \theta} \right) + \frac{R\Theta}{r^2 \sin^2\theta} \frac{\partial^2 \Phi}{\partial \phi^2} + \frac{2\mu}{\hbar^2} [E - V(r)] R\Theta\Phi = 0$$

multiplying above equation on both sides with $\frac{1}{R\Theta\Phi}$ we get;

$$\frac{1}{Rr^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial R}{\partial r} \right) + \frac{1}{\Theta r^2 \sin\theta} \frac{\partial}{\partial \theta} \left(\sin\theta \frac{\partial \Theta}{\partial \theta} \right) + \frac{1}{\Phi r^2 \sin^2\theta} \frac{\partial^2 \Phi}{\partial \phi^2} + \frac{2\mu}{\hbar^2} [E - V(r)] = 0$$

multiplying above equation with 'r²',

$$\text{or, } \frac{1}{R} \frac{\partial}{\partial r} \left(r^2 \frac{\partial R}{\partial r} \right) + \frac{1}{\Theta \sin\theta} \frac{\partial}{\partial \theta} \left(\sin\theta \frac{\partial \Theta}{\partial \theta} \right) + \frac{1}{\Phi \sin^2\theta} \frac{\partial^2 \Phi}{\partial \phi^2} + \frac{2\mu r^2}{\hbar^2} [E - V(r)] = 0$$

$$\text{or, } \frac{1}{R} \frac{\partial}{\partial r} \left(r^2 \frac{\partial R}{\partial r} \right) + \frac{2\mu r^2}{\hbar^2} [E - V(r)] = -\frac{1}{\Theta \sin\theta} \frac{\partial}{\partial \theta} \left(\sin\theta \frac{\partial \Theta}{\partial \theta} \right) - \frac{1}{\Phi \sin^2\theta} \frac{\partial^2 \Phi}{\partial \phi^2} = \lambda(\text{say})$$

Then either,

$$\frac{1}{R} \frac{\partial}{\partial r} \left(r^2 \frac{\partial R}{\partial r} \right) + \frac{2\mu r^2}{\hbar^2} [E - V(r)] = \lambda(\text{say}) \dots\dots\dots 9$$

$$\text{OR } \frac{1}{R} \frac{\partial}{\partial r} \left(r^2 \frac{\partial R}{\partial r} \right) + \frac{2\mu r^2}{\hbar^2} [E - V(r)] = -\frac{1}{\Theta \sin\theta} \frac{\partial}{\partial \theta} \left(\sin\theta \frac{\partial \Theta}{\partial \theta} \right) - \frac{1}{\Phi \sin^2\theta} \frac{\partial^2 \Phi}{\partial \phi^2} = \lambda(\text{say}) \dots\dots\dots 10$$

From equation 9; multiplying both sides with $\frac{R}{r^2}$, we get;

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial R}{\partial r} \right) + \frac{2\mu}{\hbar^2} [E - V(r)] R = \frac{\lambda R}{r^2}$$

$$\text{or, } \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial R}{\partial r} \right) + \left(\frac{2\mu}{\hbar^2} [E - V(r)] - \frac{\lambda}{r^2} \right) R = 0 \dots\dots\dots 11$$

This equation relates only the radial part of Schrodinger wave equation of an H-atom and gives radial solution so named a radial equation of H-atom. The solution of this radial equation gives the energy of an electron in H-atom in nth energy level as,

$$E_n = -\frac{me^2}{8\epsilon_0^2 \hbar^2 n^2}$$

which is same as obtained from Bohr's Theory of H-atom. Here, n = 1, 2, 3, 4, 5,..... and represent K, L, M, N shells of H-atom and hence further shows energy level for an electron in H-atom is quantized. 'n' refers the principle quantum number and has its values $n \geq l$ where l is orbital quantum number of an electron.

Further, from equation 10,

$$-\frac{1}{\Theta \sin\theta} \frac{\partial}{\partial \theta} \left(\sin\theta \frac{\partial \Theta}{\partial \theta} \right) - \frac{1}{\Phi \sin^2\theta} \frac{\partial^2 \Phi}{\partial \phi^2} = \lambda(\text{say})$$

$$\text{or, } \frac{1}{\Theta \sin\theta} \frac{\partial}{\partial \theta} \left(\sin\theta \frac{\partial \Theta}{\partial \theta} \right) + \frac{1}{\Phi \sin^2\theta} \frac{\partial^2 \Phi}{\partial \phi^2} + \lambda = 0 \dots\dots\dots 12$$

As this equation relate two angles and contains angular part of H-atom only so named polar equation. The solution of this equation for Θ gives the value of orbital quantum number l whose values is $l = 0, 1, 2, 3, \dots\dots\dots (n-1)$ with 'n' principle quantum number. This quantum number gives s, p, d, f, sub shell of atom.

multiplying both sides with 'sin²θ', we have;

$$\frac{\sin\theta}{\Theta} \frac{\partial}{\partial \theta} \left(\sin\theta \frac{\partial \Theta}{\partial \theta} \right) + \frac{1}{\Phi} \frac{\partial^2 \Phi}{\partial \phi^2} + \lambda \sin^2\theta = 0$$

$$\text{or, } \frac{\sin\theta}{\Theta} \frac{\partial}{\partial \theta} \left(\sin\theta \frac{\partial \Theta}{\partial \theta} \right) + \lambda \sin^2\theta = -\frac{1}{\Phi} \frac{\partial^2 \Phi}{\partial \phi^2}$$

$$\text{or, } \frac{\sin\theta}{\Theta} \frac{\partial}{\partial \theta} \left(\sin\theta \frac{\partial \Theta}{\partial \theta} \right) + \lambda \sin^2\theta = -\frac{1}{\Phi} \frac{\partial^2 \Phi}{\partial \phi^2} = m_l^2(\text{say})$$

Again,

$$\text{Either, } \frac{\sin\theta}{\Theta} \frac{\partial}{\partial \theta} \left(\sin\theta \frac{\partial \Theta}{\partial \theta} \right) + \lambda \sin^2\theta = m_l^2 \dots\dots\dots 13$$

$$\text{OR } -\frac{1}{\Phi} \frac{\partial^2 \Phi}{\partial \phi^2} = m_l^2 \dots\dots\dots 14 \quad \text{which is named as azimuthal wave equation.}$$

Further, from equation 14;

$$\frac{\partial^2 \Phi}{\partial \phi^2} = -m_l^2 \Phi$$

$$\text{or, } \frac{\partial^2 \Phi}{\partial \phi^2} + m_l^2 \Phi = 0 \dots\dots\dots 15$$

Which is second order differential equation in 'Φ' whose solution is;

$$\Phi(\phi) = Ae^{\pm im_l \phi} \dots\dots\dots 16$$

Since, wave function Ψ must be single value so,

$$\Phi(0) = \Phi(2\pi)$$

$$\text{or, } Ae^{\pm im_l 0} = Ae^{\pm im_l 2\pi}$$

$$\text{or, } e^{\pm im_l 2\pi} = 1$$

$$\text{or, } \cos(2\pi m_l) \pm i \sin(2\pi m_l) = 1$$

$$\text{So, real part, } \cos(2\pi m_l) = 1$$

$$\text{or, } \cos(2\pi m_l) = \cos(\pm 2n\pi) \text{ where n is integer}$$

$$\text{or, } 2\pi m_l = \pm 2n\pi$$

$$\text{or, } m_l = \pm n \quad \text{where n is integer number}$$

$$\text{So, } m_l = 0, \pm 1, \pm 2, \pm 3, \dots\dots\dots \pm l \dots\dots\dots 17$$

m_l has $(2l + 1)^{th}$ number of values for given l including zero.

The 'm' refer to the magnetic quantum number and 'l' is orbital quantum number. This shows, the magnetic quantum number in case of H-atom is quantized for given orbital quantum number and the azimuthal wave function is

$$\text{from eqn 16; } \Phi(\phi) = Ae^{\pm im_l \phi}$$

Now applying the condition of normalization above equation;

$$\int_0^{2\pi} \Phi^*(\phi) \Phi(\phi) d\phi = 1$$

$$\text{or, } \int_0^{2\pi} A e^{\mp im_l \phi} A e^{\pm im_l \phi} d\phi = 1$$

$$\text{or, } A^2 \int_0^{2\pi} d\phi = 1$$

$$\text{or, } A^2 2\pi = 1$$

$$\text{or, } A = \frac{1}{\sqrt{2\pi}}$$

So, normalized azimuthal wave function for an electron in H-atom is;

$$\Phi(\phi) = \frac{1}{\sqrt{2\pi}} e^{\pm im_l \phi} \dots\dots\dots 18$$

Physical Significance of Various Quantum Numbers: -

On solving the time independent Schrodinger wave equation by separation variable method for the wave function of an electron in H-atom we encounter a radial wave equation and polar wave equation. The solution of these equations further gives us discrete values of variable n, l, and m_l representing the quantum number in quantum mechanics. These numbers are essentially useful to describe or locate an electron around H-atom in any orbit (state) which are as describe below;

1. Interpretation of Principle quantum number(n):-

The solution of radial part of Schrodinger wave equation gives the energy levels values for an electron in H-atom. i.e.

$$E_n = -\frac{me^2}{8\epsilon_0^2 h^2 n^2}$$

Here, $n = 1, 2, 3, 4, 5, \dots$ this clearly, shows energy level for an electron in H-atom is quantized that is to say have only discrete energy value. 'n' refers here as the principal quantum number and represents K, L, M and N shell in an atom. Its values is $n \geq l$ where l is orbital quantum number of an electron.

2. Interpretation of Orbital quantum number (l):-

The solution of polar part of wave equation results to the orbital quantum number ' l '. Its values $l = 0, 1, 2, 3, \dots, (n-1)$ with 'n' principle quantum number and represent the sub shells s, p, d and f in an atom. This is further can be relate the magnitude of angular momentum of an electron in orbital quantum number as $L = \hbar\sqrt{l(l+1)}$. Thus, quantum number also clearly shows the quantization of angular momentum value of an electron in different energy levels.

3. Interpretation of magnetic Orbital quantum number (m_l):-

This quantum number is also results out from solving polar part of wave equation. This quantum number determine the orientation of angular momentum 'L' in a presence of magnetic field.

When an atom is placed in magnetic field 'B' along z-axis then the z-component of angular momentum of an electron of atom in particular orbit has a discrete values given by,

$L_z = \hbar m_l$ where m_l is magnetic orbital quantum number and takes the values $m_l = 0, \pm 1, \pm 2, \pm 3, \dots, \pm l$ with ' l ' be orbital quantum number. m_l has $(2l + 1)^{th}$ number of values for given l including zero.

This is also known as space quantization.

For, e.g. for $l = 2$; $m_l = 2, 1, 0, -1, -2$

Thus, $L_z = 2\hbar; L_z = \hbar; L_z = 0; L_z = -\hbar; L_z = -2\hbar$ as shown in figure.

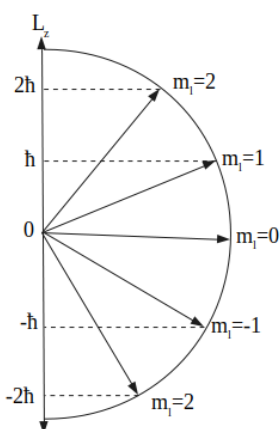


Fig: Possible orientation of quantized values of angular momentum for an electron in H-atom for $l=2$ in space.

Degeneracy of Energy Levels of H-atom:-

The phenomenon in which a number of different quantum state are having same energy state is called degeneracy and the energy level that contains more than one energy level is named as degenerate energy level.

The energy eigen value of H-atom is given by;

$$E_n = -\frac{me^2}{8\epsilon_0^2 h^2 n^2}$$

This shows energy of H-atom depends on principal quantum number 'n'. The principle quantum number further depends on orbital quantum numbers with values $l = 0, 1, 2, 3, \dots, (n-1)$ and this further have possible values of magnetic quantum number m_l with values $m_l = 0, \pm 1, \pm 2, \pm 3, \dots, \pm l$. Hence a given energy level with particular value of principle quantum number have a number of states with same energy, this is to say has a degeneracy. The total number of different quantum states in given energy level is called degree of degeneracy of energy level. The different quantum states are represented by Ψ_{n,l,m_l} .

For e.g. for K and L shell of H-atom,

for $n=1$ $l=0$; $m_l=0$ $\Psi_{1,0,0}$ one energy level

for $n=2$ $l=0$; $m_l=0$ $\Psi_{2,0,0}$

$l=1$; $m_l=1, 0, -1$ $\Psi_{2,1,1}, \Psi_{2,1,0}, \Psi_{2,1,-1}$ 4 degenerate energy levels for $n=2$

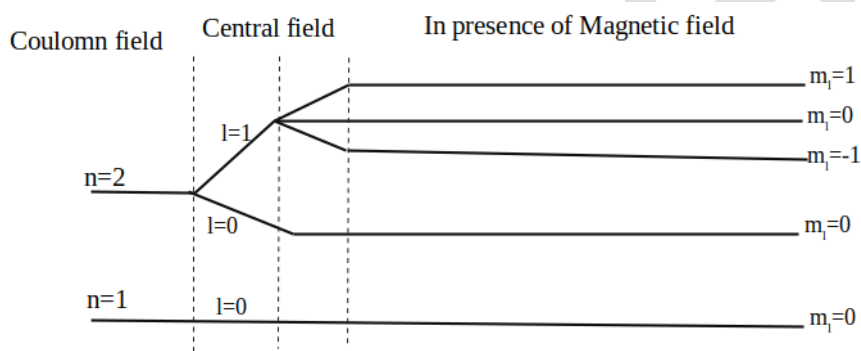


Fig: The degeneracy of energy levels of H-atom

Space quantization Experiments:

The orientation of position of an electron in terms of its angular momentum in the presence of external magnetic field is called space quantization. The concept of space quantization provides the explanation of zeeman effects and stark effect.

Zeeman Effect Experiment: -

The splitting of spectral lines of any atom into three distinct spectrum or components in the presence of uniform strong external magnetic field is called Zeeman effect. It is of two types anomalous Zeeman effect and normal Zeeman effect.

Let, us consider an atom in which an electron is revolving around the nucleus in a circular orbit 'r' with linear velocity 'v'. As electron possess 'e' charge and hence results to flow of I current in a close loop of area 'A'. The magnetic dipole moment due to motion of electron in 'l' orbital motion is,

$$\mu_l = IA \dots \dots \dots 1$$

When such atom is placed in external magnetic field then, the dipole moment experience torque given by,

$$\tau = \mu_l \times B = \mu_l B \sin \theta \dots \dots \dots 2$$

The potential energy of magnetic dipole in a magnetic field is,

$$E_p = -\vec{\mu}_l \cdot \vec{B} = -\mu_l B \cos \theta \dots \dots \dots 3$$

Now, the magnetic dipole of an atom is from equation 1

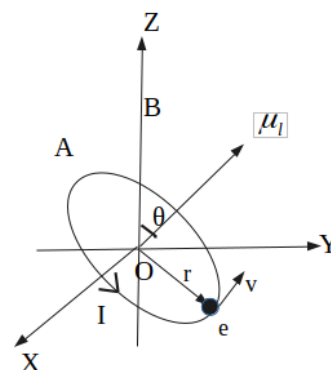


Fig: A magnetic moment in a magnetic field B.

$$\mu_l = \frac{|e|}{T} A$$

$$\text{or, } \mu_l = \frac{|e|}{\frac{2\pi r}{v}} \pi r^2$$

$$\text{or, } \mu_l = \frac{|e|vr}{2}$$

$$\text{or, } \mu_l = \frac{|e|mvr}{2m}$$

$$\text{or, } \mu_l = -\frac{|e|L}{2m} \text{ where } L=mvr \text{ is angular momentum of an electron in an orbit}$$

and -ve sign is taken showing the direction of 'L' is opposite to direction of motion of an electron.

Then from equation 3;

$$E_p = \frac{|e|L}{2m} B \cos \theta$$

$$\text{or, } E_p = \frac{|e|}{2m} B L_z \text{ here, } L_z \text{ is the component of}$$

angular momentum of electron along z-axis and

$$L_z = \hbar m_l$$

So, the potential energy of a dipole in a magnetic field is,

$$E_p = \frac{|e|B\hbar m_l}{2m} \dots\dots\dots 4$$

Therefore, the energy of an electron in given orbit when an atom is placed in external magnetic field is;

$$E_T = E_n + E_p$$

$$\text{or, } E_T = E_n + \frac{|e|B\hbar m_l}{2m} \dots\dots\dots 5$$

As m_l takes the values $m_l = 0, \pm 1, \pm 2, \pm 3, \dots\dots\dots \pm l$ with given value of orbital quantum number 'l' and principle quantum number 'n' and hence given energy level splits into a number of sub levels i.e. degenerate energy levels.

The transition of an electron from higher energy level to lower energy level is followed by a certain rule named selection rule and electron following this gives rise a emission spectral lines. According to selection rule, those transitions are only possible that has value $\Delta m_l = 0, \pm 1$.

Hence, the energy difference between two energy levels is given by;

$$\Delta E = \frac{|e|B\hbar}{2m} \Delta m_l$$

$$\text{or, } |\Delta E| = \frac{|e|B\hbar}{2m} \dots\dots\dots 6 \quad [\text{ as } \Delta m_l = \pm 1 \text{ selection rule }]$$

Further,

As $E = h\nu = \frac{hc}{\lambda}$ and on differentiating this equation with respect to 'λ' ; we have

$$dE = \frac{-hc}{\lambda^2} d\lambda \quad \text{i.e. } \Delta E = \frac{-hc}{\lambda^2} \Delta \lambda$$

Then from equation 6;

$$\left| \frac{-hc}{\lambda^2} \Delta \lambda \right| = \frac{|e|B\hbar}{2m}$$

$$\text{or, } |\Delta \lambda| = \frac{|e|\lambda^2 B \hbar}{2mhc}$$

$$\text{or, } |\Delta \lambda| = \frac{B|e|\lambda^2}{2mhc} \frac{h}{2\pi}$$

$$\text{or, } |\Delta \lambda| = \frac{B|e|\lambda^2}{4\pi mc} \dots\dots\dots 7$$

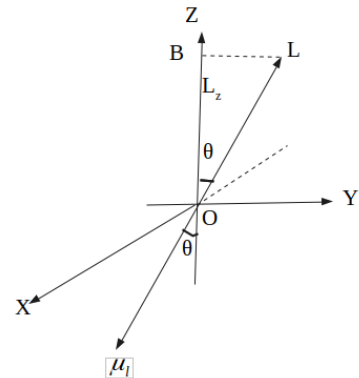


Fig: Orientation of angular momentum and magnetic diople moment of an electron in presence of an external magnetic field.

Note: $|\Delta\lambda| = \nu \frac{\lambda^2}{c}$ where, $\nu = \frac{B|e|}{4\pi m}$ is precession frequency of an electron about an axis of applied magnetic field also called Larmor frequency.

This gives the change in wave lengths of spectrum or separation between three components of normal Zeeman pattern.

For e.g the transition of an electron from $l=2$ to $l=1$ for given principle quantum number $n=3$. The possible transitions under selection rule results to nine spectral lines with three different energy values as shown in figure.

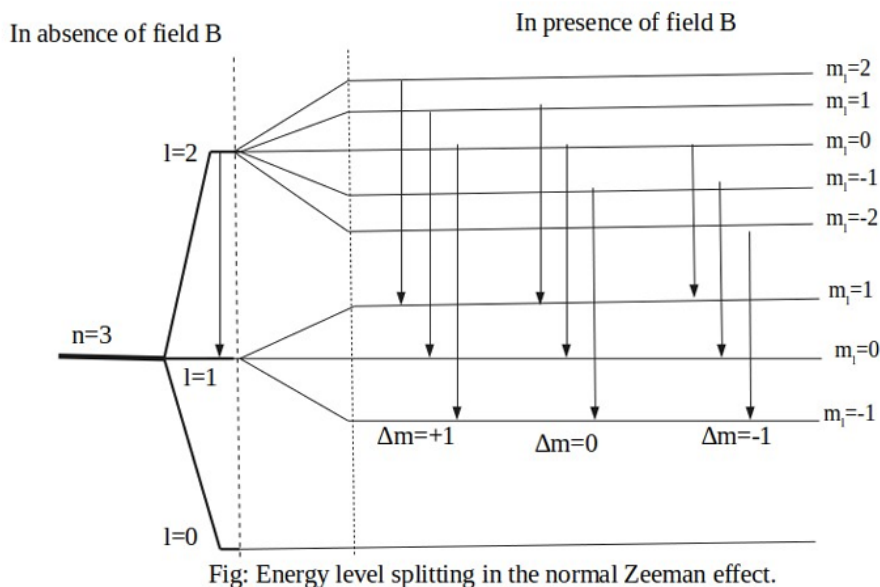


Fig: Energy level splitting in the normal Zeeman effect.

Thus, such splitting of energy levels in presence of uniform strong magnetic field is known as Normal Zeeman effect. This type of effect was experimentally observed in case of helium, calcium atoms and thus shows the idea of space quantization.

Stern Gerlach Experiment:

In 1921, Otto stern and Walter Gerlach performed an experiment and prove that the existence of electron spin and space quantization. They demonstrated that the spatial orientation of angular momentum is quantized into two components up and down.

In the experiment, Stern and Gerlach sent a beam of neutral silver atoms obtained by heating silver at oven through the collimators and then through a non-uniform magnetic field region along z-axis perpendicularly. Finally, the atoms at screen are obtained to be striking at three different spot on photographic film as shown in figure.

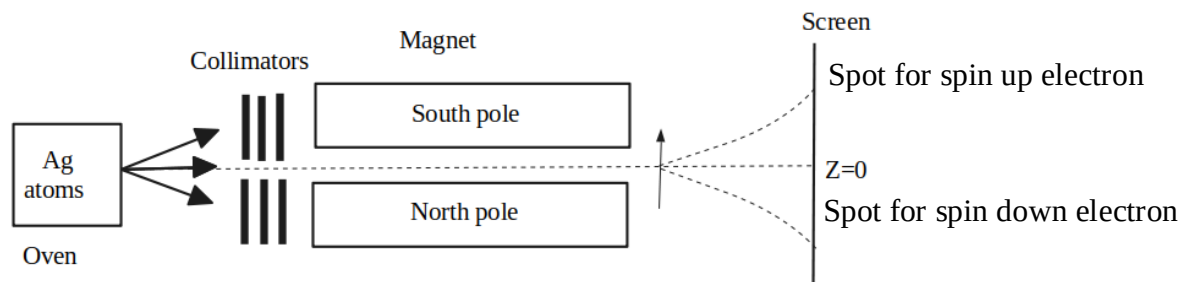


Fig: Diagrammatic representation of experimental set up of Stern Gerlach experiment.

A high value of non-uniform magnetic field is produced by using powerful electromagnet with a knife edge shape. The magnetic field is much stronger at edge and reduced to others region of pole pieces.

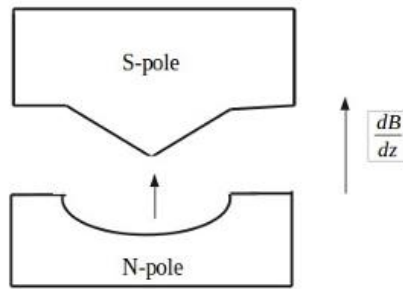


Fig: Edge shape pole piece producing inhomogeneous magnetic field.

When an atom or a magnetic dipole is placed in a non-uniform magnetic field it experiences force however in uniform magnetic field it experiences a torque.

Let, $\frac{dB}{dz}$ be the magnetic gradient along z-axis in which an electron having spin magnetic dipole of dipole moment value μ_s is placed and for θ be the angle between the dipole and non uniform magnetic field. Then, the dipole experiences a force whose magnitude along z- axis is,

$$F_z = \frac{-dU}{dz}$$

$$\text{or, } F_z = \frac{d(\mu_s B \cos\theta)}{dz}$$

$$\text{or, } F_z = \mu_s \cdot \cos\theta \frac{dB}{dz} \dots\dots\dots 1$$

Now, the magnetic dipole of an atom due to an electron 'e' spin is,

$$\mu_s = IA$$

$$\text{or, } \mu_s = \frac{|e|\hbar}{2m} A$$

$$\text{or, } \mu_s = \frac{|e|\hbar}{2m} \pi r^2 v$$

$$\text{or, } \mu_s = \frac{|e|\hbar v r}{2}$$

$$\text{or, } \mu_s = \frac{|e|\hbar m v r}{2m}$$

or, $\mu_s = -\frac{|e|\hbar L_s}{2m} \dots\dots\dots 2$ where $L_s = mvr$ is angular momentum of an electron due to spin and -ve sign is taken showing the direction of ' L_s ' is opposite to direction of μ_s . Thus obtain magnetic moment due to spin of electron donot match with experimental value and so a correction factor with $g=2$ (landu g-factor) is introduced. Thus

$$\mu_s = -g \frac{|e|\hbar L_s}{2m}$$

$$\text{or, } \mu_s = -\frac{|e|\hbar L_s}{m} \dots\dots\dots 3$$

From 1 and 3, we get;

$$F_z = \mu_s \cdot \cos\theta \frac{dB}{dz}$$

$$\text{or, } F_z = -\frac{|e|\hbar}{m} L_s \cos\theta \frac{dB}{dz}$$

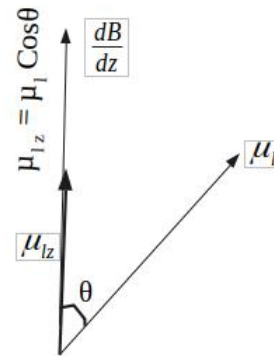


Fig: orientation of magnetic dipole along non uniform magnetic field direction.

or, $F_z = -\frac{|e|\hbar}{m} L_{sz} \frac{dB}{dz}$ 3 where $L_{sz} = L_s \cos\theta$ is component of spin angular momentum along z-axis and its value is $L_{sz} = \hbar m_s$. Here, $m_s = \pm \frac{1}{2}$ is magnetic spin quantum number.

Thus, $F_z = -\frac{|e|\hbar}{m} m_s \frac{dB}{dz}$

Or, $F_z = \pm \frac{|e|\hbar}{2m} \frac{dB}{dz}$

So, the magnetic dipoles experiences a forces in z-axis that can have value between $\frac{|e|\hbar}{2m} \frac{dB}{dz}$ to $-\frac{|e|\hbar}{2m} \frac{dB}{dz}$ and obtained two spot up and down on screen at equal distance in the experimentlly .Thus, in this way Stern and Gerlach experiement gives the idea of electrons are spin $\frac{1}{2}$ particles and their spin angular momentum in space is quantized.

Spin of an electron: -

in 1925, S. Goudsmit and G. Uhlenbeck proposed that the electron has an intrinsic angular momentum due to spinning motion about their axis of rotation. The magnitude of spinning of an electron is measured in term of corresponding angular momentum 'S' and spin dipole moment ' μ_s '. The value of 'S' is quantized along z-axis in presence of external magnetic field is along this axis and is given by;

$$S = \hbar \sqrt{s(s+1)} \quad \text{and } s = \frac{1}{2} \text{ is spin value of an electron.}$$

Further, magnitude of magnetic Spin dipole moment is given by; $m_s = \pm \frac{1}{2}$

Hence, in order to specify the state of an electron in quantum mechanics about an atom, it is required all together four fundamental quantum numbers

1. Principle quantum number $n=1,2,3,4,\dots$
2. Orbital quantum number $l=0, 1, 2, 3, \dots (n-1)$
3. Magnetic orbital quantum number $m_l = 0, \pm 1, \pm 2, \pm 3, \pm 4, \dots, \pm l$
4. Magnetic spin quantum number $m_s = \pm \frac{1}{2}$